

Path-spread-resolved quantum transition-state theory for proton transfer

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Centroid quantum transition-state theory and ring-polymer molecular dynamics yield thermal rates, but they collapse the imaginary-time path of the transferring proton onto a single collective coordinate. Distinct compact and delocalized ring-polymer geometries that share the same centroid are therefore averaged together, and the tunneling character of the reactive flux is not resolved. We define the proton-path spread χ_p as the mean-square bead displacement of the ring polymer along the proton-sharing coordinate—equivalently, the squared radius of gyration projected onto that coordinate—and condition the quantum transition-state flux on χ_p . The resulting path-spread-resolved (QRR) estimator is a decomposition of the usual centroid flux, not a competing dynamical method. For a harmonic proton mode, $\langle\chi_p\rangle$ has a closed continuum formula and an exact finite-bead counterpart; Levy-staging path-integral Monte Carlo reproduces the finite-bead result to within 1%. The H/D ratio of $\langle\chi_p\rangle$ interpolates between $\sqrt{m_D/m_H}$ in the deep-quantum limit and m_D/m_H in the Wigner regime. Exact Eckart thermal rates at 300 K show that the tunneling correction κ_H grows from 1.28 to 2.68 as the barrier increases from 4 to 16 $k_B T$, while the kinetic isotope effect rises from 1.68 to 2.28 against a classical value of $\sqrt{2}$. On an asymmetric double well, umbrella-sampled free-energy surfaces $F(\tilde{\xi}_p, \chi_p)$ resolve compact and delocalized contributions at the dividing surface as a function of barrier height and isotope. A two-dimensional proton-solvent model supplies a numerical factorization diagnostic based on $\text{corr}(\chi_p, q_s)$ at the dividing surface. The construction is an add-on collective variable for any existing path-integral code.

I. INTRODUCTION

Proton transfer, and proton-coupled electron transfer (PCET) in particular, is a setting in which nuclear quantum effects, solvent reorganization, and, at electrodes, the applied potential jointly control the rate.^{1–4} Kinetic isotope effects (KIEs) and non-Arrhenius temperature dependence are the usual experimental signatures of tunneling,^{4,5} but they do not by themselves identify the imaginary-time path geometry that carries the reactive flux.

Path-integral methods address that geometry at the level of the thermal density matrix.^{6–8} Centroid quantum transition-state theory (QTST) replaces the classical reaction coordinate by the ring-polymer centroid and evaluates a flux through a dividing surface in centroid space.^{9–12} Ring-polymer molecular dynamics (RPMD) propagates the same polymer in real time and supplies a recrossing correction to the centroid flux.^{13–15} Instanton theory, in the complementary deep-tunneling limit, optimizes a single dominant imaginary-time orbit.^{16,17} Transition-path sampling retains full real-time path information, at a cost that is not organized by equilibrium imaginary-time geometry.¹⁸

What these approaches do not provide is a cheap, equilibrium order parameter that distinguishes compact ring polymers (thermally activated, localized near the donor

or acceptor) from delocalized polymers whose beads span the transfer coordinate (tunneling-like imaginary-time geometry). The ring-polymer radius of gyration is already used as a diagnostic of delocalization in condensed-phase proton studies.^{19,20} Here we restrict that diagnostic to the chemical sharing coordinate, give it exact harmonic limits, and use it to *condition* the QTST flux.

The resulting path-spread-resolved rate, which we call QRR, is

$$k_{\text{QRR}} = \int_0^\infty d\chi_p k_{\text{QTST}}(\chi_p) \kappa(\chi_p) p^\ddagger(\chi_p), \quad (1)$$

where $p^\ddagger(\chi_p)$ is the distribution of path spreads at the dividing surface and $\kappa(\chi_p)$ is a class-conditional transmission coefficient. When κ is taken independent of χ_p , Eq. (1) reduces identically to ordinary QTST. QRR is therefore a decomposition of a known estimator, not a claim of a more accurate thermal rate. Its value is mechanistic: it reports which imaginary-time geometries carry the flux, how that mix depends on isotope and barrier, and whether path spread couples to a solvent coordinate.

Section II defines χ_p and the conditional free energy, states the solvable limits, and writes the flux decomposition. Section III describes the samplers. Section IV reports harmonic, Eckart, double-well, and coupled-solvent calculations. Section V discusses scope and limitations. We do not report condensed-phase electrochemical simulations; those require a validated reactive potential and are left as an application of the estimator.

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II. THEORY

A. Path-spread functional

The P -bead ring-polymer representation of the canonical density is standard.^{8,10} For a proton transferring between donor and acceptor oxygens we use the bead-resolved sharing coordinate

$$\xi_p^{(s)} = r_{\text{ODH}}^{(s)} - r_{\text{OAH}}^{(s)}, \quad \bar{\xi}_p = P^{-1} \sum_{s=1}^P \xi_p^{(s)}. \quad (2)$$

The path-spread functional is the bead variance along that coordinate,

$$\chi_p = \frac{1}{P} \sum_{s=1}^P (\xi_p^{(s)} - \bar{\xi}_p)^2. \quad (3)$$

This is the squared radius of gyration of the polymer projected onto ξ_p . It is identically zero at $P = 1$ and is non-negative for all configurations. Compact paths have χ_p of order the zero-point spread in a well; delocalized paths have χ_p large enough that beads occupy both sides of the barrier.

The joint density $P(\bar{\xi}_p, \chi_p)$ defines the conditional free energy

$$F(\bar{\xi}_p, \chi_p) = -k_B T \ln P(\bar{\xi}_p, \chi_p) + C. \quad (4)$$

Projection onto $\bar{\xi}_p$ recovers the centroid potential of mean force.¹⁰ Slices at fixed $\bar{\xi}_p$ resolve path class, as sketched in Fig. 1. A solvent or energy-gap coordinate λ_s may be added without changing the construction; the three-dimensional surface $F(\bar{\xi}_p, \chi_p, \lambda_s)$ is the natural object for PCET, but the calculations below are two-dimensional unless stated.

B. Flux decomposition

Let $\bar{\xi}_p^\ddagger$ locate the dividing surface. The marginal

$$p^\ddagger(\chi_p) = \frac{e^{-\beta F(\bar{\xi}_p^\ddagger, \chi_p)}}{\int d\chi_p' e^{-\beta F(\bar{\xi}_p^\ddagger, \chi_p')}} \quad (5)$$

is the probability that a dividing-surface configuration belongs to path class χ_p . The class-conditional QTST flux is the usual positive-velocity centroid flux restricted to that class,

$$k_{\text{QTST}}(\chi_p) = Z_R^{-1} \langle [\dot{\bar{\xi}}_p] + \delta(\bar{\xi}_p - \bar{\xi}_p^\ddagger) \delta(\chi_p[\mathbf{R}] - \chi_p) \rangle. \quad (6)$$

A Bennett–Chandler transmission coefficient $\kappa(\chi_p)$ may be attached to each class.^{14,21,22} Integrating over χ_p produces Eq. (1). We emphasize two restrictions. First, $\kappa(\chi_p)$ computed from a Wigner series in $\hbar^2$ is a tunneling correction to a parabolic barrier,^{23,24} not a recrossing

factor; the two must not be conflated. Second, for highly delocalized paths the Wigner series is not a controlled approximation, and a class-conditional RPMD flux-side correlation is the consistent dynamical correction.¹⁴ The numerical work below reports $p^\ddagger(\chi_p)$ and related fractions, which are equilibrium properties, and uses exact Eckart rates when a thermal rate is required.

C. Solvable limits

Theorem 1 (Classical limit) *At $P = 1$, $\chi_p = 0$ for every configuration, $p^\ddagger(\chi_p) = \delta(\chi_p)$, and k_{QRR} reduces to classical transition-state theory.*

This is immediate from Eq. (3). It is recorded because any numerical implementation must reproduce it.

Theorem 2 (Harmonic continuum) *For $V = \frac{1}{2}m\omega^2q^2$,*

$$\langle \chi_p \rangle_\infty = \frac{\hbar}{2m\omega} \coth\left(\frac{\beta\hbar\omega}{2}\right) - \frac{k_B T}{m\omega^2}. \quad (7)$$

The first term is the equal-time imaginary-time correlator $G(0)$; the second is the centroid variance, which is classical for a harmonic oscillator.¹⁰ The difference is non-negative because $\coth x \geq 1/x$ for $x > 0$. As $T \rightarrow 0$, $\langle \chi_p \rangle \rightarrow \hbar/(2m\omega)$. As $T \rightarrow \infty$, the leading term is the Wigner correction $\hbar^2\beta/(12m)$, independent of ω .²³ The Matsubara-mode derivation is given in the supplementary material.

Proposition 1 (Exact finite P) *For the same oscillator discretized with P beads,*

$$\langle \chi_p \rangle_P = \frac{1}{P} \sum_{k=1}^{P-1} \frac{1}{\beta\lambda_k}, \quad \lambda_k = \frac{4mP}{\hbar^2\beta^2} \sin^2\left(\frac{\pi k}{P}\right) + \frac{m\omega^2}{P}. \quad (8)$$

As $P \rightarrow \infty$, $\langle \chi_p \rangle_P \rightarrow \langle \chi_p \rangle_\infty$.

Equation (8) is the quantity that a primitive path-integral estimator of Eq. (3) actually samples. Comparing Monte Carlo to Eq. (7) at modest P therefore mixes statistical error with Trotter error. We compare to Eq. (8) first.

Theorem 3 (Harmonic isotope scaling) *On a shared harmonic surface ($k = m\omega^2$ fixed), $\langle \chi_p \rangle_H > \langle \chi_p \rangle_D$ for all $T > 0$. The ratio $\mathcal{R} = \langle \chi_p \rangle_H / \langle \chi_p \rangle_D$ satisfies*

$$\mathcal{R}(T \rightarrow 0) = \sqrt{m_D/m_H}, \quad \mathcal{R}(T \rightarrow \infty) = m_D/m_H. \quad (9)$$

The low-temperature limit is the ratio of zero-point amplitudes. The high-temperature limit is the ratio of leading Wigner corrections. Both statements use Eq. (7) and do not extend automatically to anharmonic barriers; they are tests of the estimator, not predictions for condensed-phase PCET.

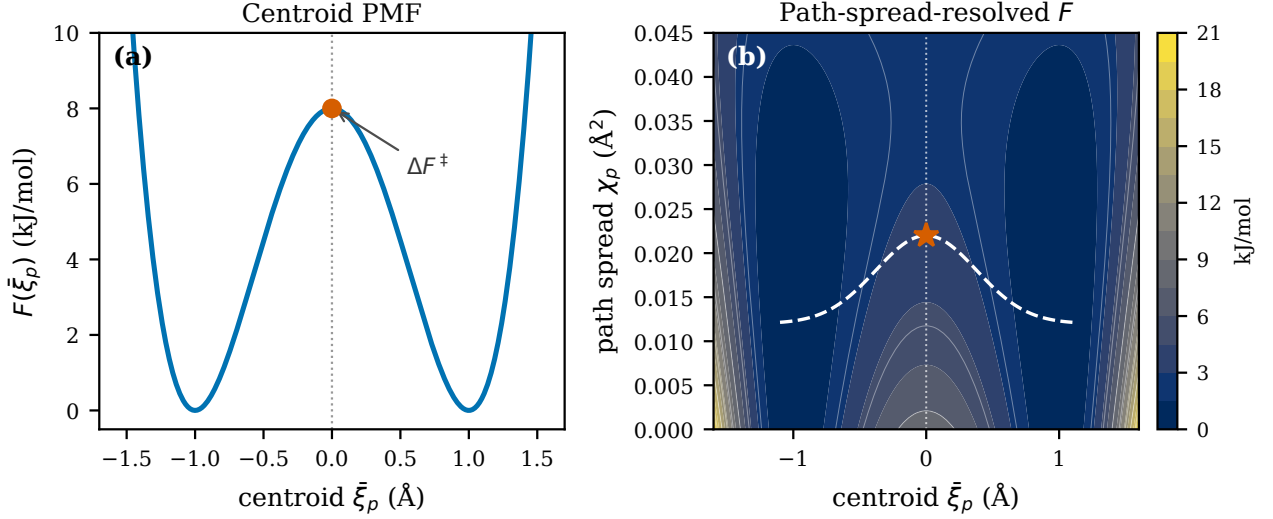


FIG. 1. (a) Centroid potential of mean force $F(\bar{\xi}_p)$, in which all ring-polymer shapes at a given centroid are averaged. (b) Conditional free energy $F(\bar{\xi}_p, \chi_p)$. The dashed curve is a schematic minimum-free-energy path; the star marks the dividing-surface path spread.

Theorem 4 (Factorization) *If $F(\bar{\xi}_p, \chi_p, \lambda_s) = F_p(\bar{\xi}_p, \chi_p) + F_s(\lambda_s)$, then $p^\ddagger(\chi_p)$ is independent of λ_s and the χ_p integral in Eq. (1) reduces to ordinary QTST when κ is class-independent.*

The mixed derivative $\partial^2 F / \partial \chi_p \partial \lambda_s$, or the dividing-surface correlation $\text{corr}(\chi_p, \lambda_s)$, is then a numerical diagnostic of proton-solvent quantum reorganization beyond a product Marcus picture.^{2,3}

III. METHODS

Path-integral Monte Carlo uses Levy staging of the free-particle bridge,^{25,26} which samples the ring-polymer springs exactly, together with a Metropolis decision on the physical potential and a rigid centroid move. The primitive estimator of χ_p is the bead variance. For the harmonic oscillator we also evaluate Eqs. (7) and (8) in closed form.

Asymmetric double-well (ADW) free energies are obtained from one-dimensional umbrella sampling along $\bar{\xi}_p$ ²⁷ with harmonic restraints, reweighted by the weighted histogram analysis method (WHAM).²⁸ Two-dimensional histograms of $(\bar{\xi}_p, \chi_p)$ use the same WHAM weights. The delocalized fraction at the dividing surface is

$$f_{\text{deloc}} = \int_{\chi_p \geq 2\chi_c} p^\ddagger(\chi_p) d\chi_p, \quad (10)$$

with the threshold $\chi_c = \hbar / (2m\omega^*)$ evaluated at $\omega^* = 1000 \text{ cm}^{-1}$, giving $\chi_c \approx 0.010 \text{ \AA}^2$. Compact and intermediate classes use $0.5\chi_c$ and $2\chi_c$ as bookkeeping cuts; the continuous $p^\ddagger(\chi_p)$ is the primary object.

Exact thermal rates for the symmetric Eckart barrier $V(x) = V_0 \text{sech}^2(x/a)$ are

$$k = \frac{1}{2\pi\hbar Q_r} \int_0^\infty dE e^{-\beta E} T(E), \quad (11)$$

with $T(E)$ the Connor transmission probability²⁹ and $Q_r = (m/2\pi\hbar^2\beta)^{1/2}$. Classical TST is $k_{\text{TST}} = \sqrt{k_B T / (2\pi m)} e^{-\beta V_0}$. The Wigner estimate multiplies k_{TST} by $1 + (\beta\hbar\omega_b)^2/24$, where $\omega_b = \sqrt{2V_0/(ma^2)}$. All Eckart integrals use a logarithmically stable evaluation of $T(E)$.

A two-dimensional model

$$V(q_p, q_s) = V_{\text{ADW}}(q_p) + \frac{1}{2}k_s q_s^2 + \kappa_c q_p q_s \quad (12)$$

treats the proton as a ring polymer and the solvent coordinate as classical. The factorization diagnostic is $\text{corr}(\chi_p, q_s)$ restricted to $|\bar{\xi}_p| < 0.10 \text{ \AA}$.

Temperatures are 300 K unless noted. Masses are 1 and 2 amu. Random number generators are seeded at 42. Code and JSON outputs are in the repository accompanying this article.

IV. RESULTS

A. Harmonic oscillator

Figure 2 compares Eq. (7) with Levy-staging Monte Carlo for a 1000 cm^{-1} mode over 100–700 K. At 300 K the continuum values are $\langle \chi_p \rangle_H = 0.01011 \text{ \AA}^2$ and $\langle \chi_p \rangle_D = 0.00572 \text{ \AA}^2$. With $P = 64$ (H) and $P = 32$ (D), the exact finite-bead formula differs from the continuum by 0.12% and 0.29%, respectively, and the Monte

Carlo root-mean-square relative error against Eq. (8) is 0.56% (H) and 1.03% (D).

The isotope ratio $\mathcal{R}(T)$ increases from 1.52 at 100 K toward $\sqrt{2}$ as T is lowered, and from 1.77 at 300 K to 1.94 at 700 K, approaching 2 from below, as required by Theorem 3. Figure 3 shows that the primitive estimator tracks Eq. (8) from $P = 4$ to $P = 128$ and that the Trotter error relative to the continuum is already below 0.5% at $P = 32$ for this frequency and temperature. A staging move that draws Brownian-bridge *marginals* independently, rather than the sequential Lévy construction, produces a spurious increase of $\langle \chi_p \rangle$ with P and should not be used.

B. Eckart barrier

The symmetric Eckart barrier with width $a = 0.50 \text{ \AA}$ provides thermal rates that do not depend on path-integral sampling. Figure 4(a) shows the exact tunneling correction $\kappa = k/k_{\text{TST}}$ at 300 K. For H, κ increases from 1.28 at $V_0 = 4 k_{\text{B}}T$ to 2.68 at $V_0 = 16 k_{\text{B}}T$; the corresponding Wigner factors are 1.22 and 1.86. Wigner is therefore a useful first correction at moderate $\hbar\omega_b/k_{\text{B}}T$ and underestimates the exact κ once $\hbar\omega_b/k_{\text{B}}T \gtrsim 4$. Deuterium remains closer to TST, as expected from the lower barrier frequency.

The exact KIE [Fig. 4(b)] is 1.68 at $4 k_{\text{B}}T$ and 2.28 at $16 k_{\text{B}}T$. Classical TST contributes only $\sqrt{m_D/m_H} = \sqrt{2} \approx 1.414$, independent of barrier height. The Wigner KIE tracks the exact result qualitatively (1.55 to 1.84) but lags at the highest barriers. These numbers set the scale of the nuclear quantum effect that a path-spread decomposition is intended to resolve: when κ_H is of order two, a non-negligible fraction of the flux is expected to occupy delocalized imaginary-time paths.

C. Asymmetric double well

The ADW $V(q) = V_0[(q/q_0)^2 - 1]^2 + \alpha q$ with $q_0 = 0.70 \text{ \AA}$ is a bound proton-transfer model. Figure 5 shows WHAM-reweighted $F(\tilde{\xi}_p, \chi_p)$ for H at $V_0 = 2 k_{\text{B}}T$ and $8 k_{\text{B}}T$. The centroid dividing surface $\tilde{\xi}_p = 0$ cuts a ridge whose χ_p support is broader than in the wells. Figure 6(a) reports f_{deloc} versus V_0 . For H the delocalized fraction is 0.18 at $2 k_{\text{B}}T$ and 0.23 at $8 k_{\text{B}}T$; for D it is an order of magnitude smaller (0.014 to 0.045) and increases with barrier height. The dividing-surface mean spread [Fig. 6(b)] is $\langle \chi_p \rangle_{\ddagger}^{\text{H}} \approx 0.014\text{--}0.015 \text{ \AA}^2$ against $\langle \chi_p \rangle_{\ddagger}^{\text{D}} \approx 0.007\text{--}0.008 \text{ \AA}^2$. Centroid barriers from WHAM are lower for H than for D (4.25 versus 4.60 kJ mol^{-1} at $2 k_{\text{B}}T$; 15.9 versus 17.6 kJ mol^{-1} at $8 k_{\text{B}}T$), as expected from greater zero-point and tunneling contributions.

These are equilibrium path-class statistics, not rates. We do not convert f_{deloc} into a claimed improvement over RPMD. Combined with the Eckart KIEs of Fig. 4, they

show that the same conditions that enhance tunneling in an exact one-dimensional rate also enhance the weight of delocalized ring-polymer geometries at an ADW dividing surface, and that the isotope contrast in f_{deloc} is large.

D. Proton-solvent coupling

Theorem 4 is tested on the bilinear model at $V_0 = 4 k_{\text{B}}T$. Figure 7(a) shows $\text{corr}(\chi_p, q_s)$ at the dividing surface. At $\kappa_c = 0$ the correlation is -0.06 from 737 dividing-surface samples, consistent with noise. At $\kappa_c = 4 \text{ kJ mol}^{-1} \text{ \AA}^{-2}$ it is -0.16 , but the number of crossings drops to 320 because the bilinear term tilts the well, so the present scan does not support a precise slope of correlation versus coupling. Figure 7(b) shows f_{deloc} on the same trajectories. The practical point is that the diagnostic is a covariance of two collective variables already available in a path-integral run; a production application would accumulate it over umbrella windows rather than unbiased dynamics.

V. DISCUSSION

QRR does not replace RPMD, instanton theory, or vibronic PCET rate laws.^{2,14,17,30} RPMD remains the appropriate dynamical correction to centroid QTST in the crossover regime; instanton theory remains the appropriate semiclassical description when a single tunneling orbit dominates; Hammes-Schiffer vibronic theory remains the appropriate nonadiabatic framework when electron-proton nonadiabaticity is essential. What QRR adds is an equilibrium resolution of the imaginary-time geometries that those methods otherwise average or optimize.

Two consequences follow for condensed-phase PCET, as hypotheses rather than numerical claims. First, an increase of the observed KIE with thermodynamic driving force, reported in several electrochemical proton-discharge studies,^{30,31} should coincide with an increase of $\Delta f_{\text{deloc}} = f_{\text{deloc}}^{\text{H}} - f_{\text{deloc}}^{\text{D}}$ if the extra isotope effect is carried by delocalized paths. Second, a breakdown of $F = F_p + F_s$ at the dividing surface is a concrete, computable signature that solvent reorganization and proton delocalization are not statistically independent, which is outside the product structure of Marcus theory. Testing either statement requires a reactive potential for a specific interface or enzyme, which is not constructed here.

Limitations are sharp. The path-spread functional is a property of the imaginary-time polymer, not a real-time spectroscopic observable; neutron Compton profiles and related measurements constrain momentum distributions, not χ_p directly. The three-class threshold χ_c is a bookkeeping device tied to a reference frequency; any application should report the full $p^{\ddagger}(\chi_p)$. Class-conditional recrossing is not computed. The models are one- and two-dimensional. Finite- P errors in χ_p are under control

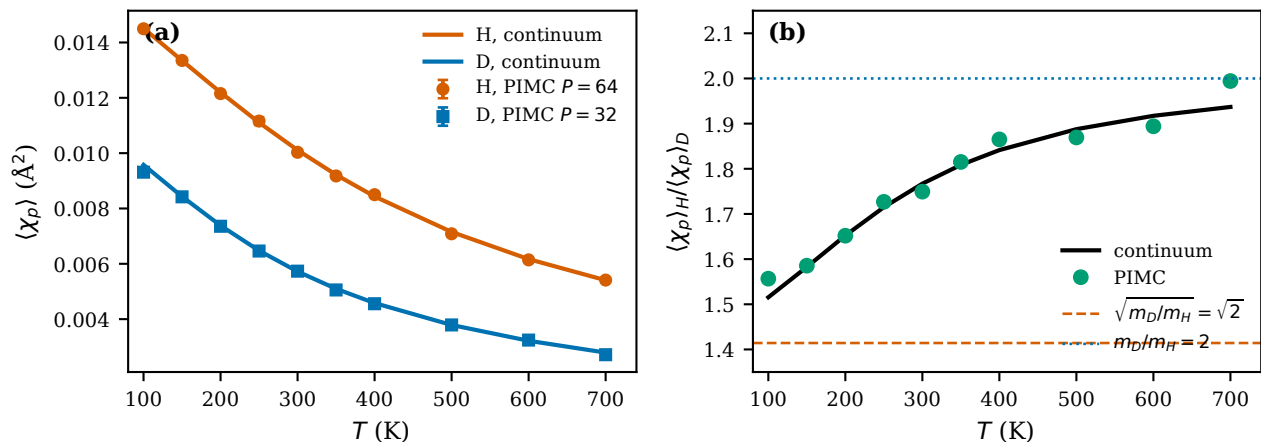


FIG. 2. Harmonic path spread for $\omega = 1000 \text{ cm}^{-1}$. (a) $\langle \chi_p \rangle(T)$ from Eq. (7) (lines) and PIMC (symbols). (b) Isotope ratio, with deep-quantum ($\sqrt{2}$) and Wigner (2) limits.

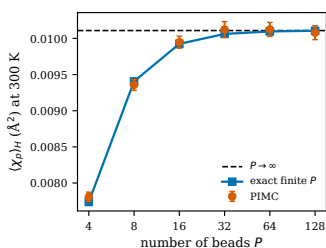


FIG. 3. Bead convergence of $\langle \chi_p \rangle_H$ at 300 K. The primitive PIMC estimator follows the exact finite- P formula, Eq. (8), and both approach the continuum, Eq. (7).

for the 1000 cm^{-1} mode at $P \geq 32$, but stiffer stretches will need more beads or a virial estimator.

VI. CONCLUSIONS

We have defined a projected ring-polymer path spread χ_p for proton transfer, derived its harmonic continuum and finite-bead limits, and used it to decompose the centroid QTST flux by imaginary-time geometry. Levy-staging Monte Carlo reproduces the exact finite-bead harmonic formula to about one percent. Exact Eckart rates quantify the tunneling corrections and KIEs that the decomposition is meant to resolve. Double-well landscapes and a coupled solvent coordinate show that delocalized-path weight and proton-solvent correlation are computable equilibrium diagnostics. The estimator is a collective variable, not a new dynamics, and can be accumulated in any path-integral molecular dynamics run.

ACKNOWLEDGMENTS

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available within the article and its supplementary material. Analysis scripts and JSON result files are included with the source distribution.

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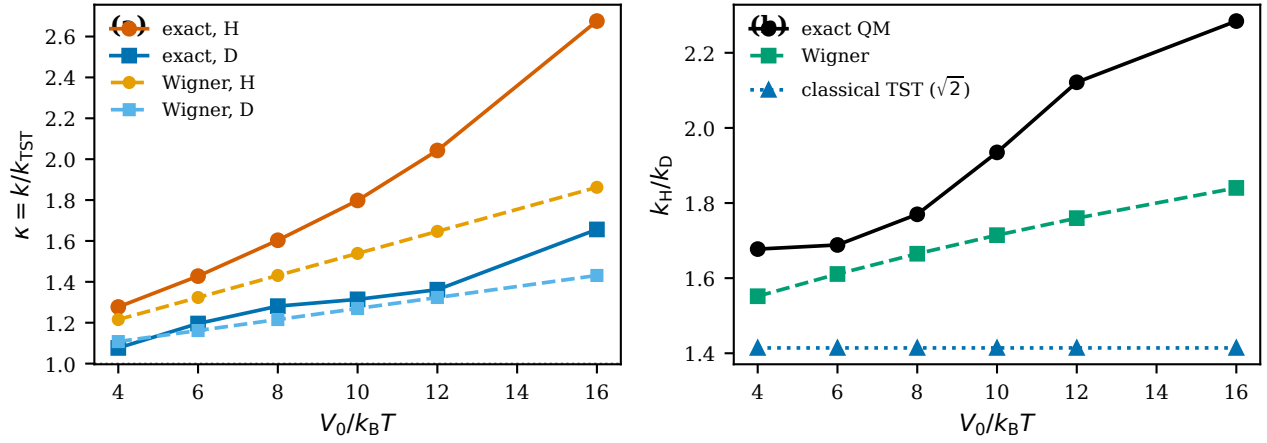


FIG. 4. Symmetric Eckart barrier, $a = 0.50 \text{ \AA}$, $T = 300 \text{ K}$. (a) Exact and Wigner tunneling corrections versus barrier height. (b) Kinetic isotope effect. Classical TST is $\sqrt{2}$ at every V_0 .

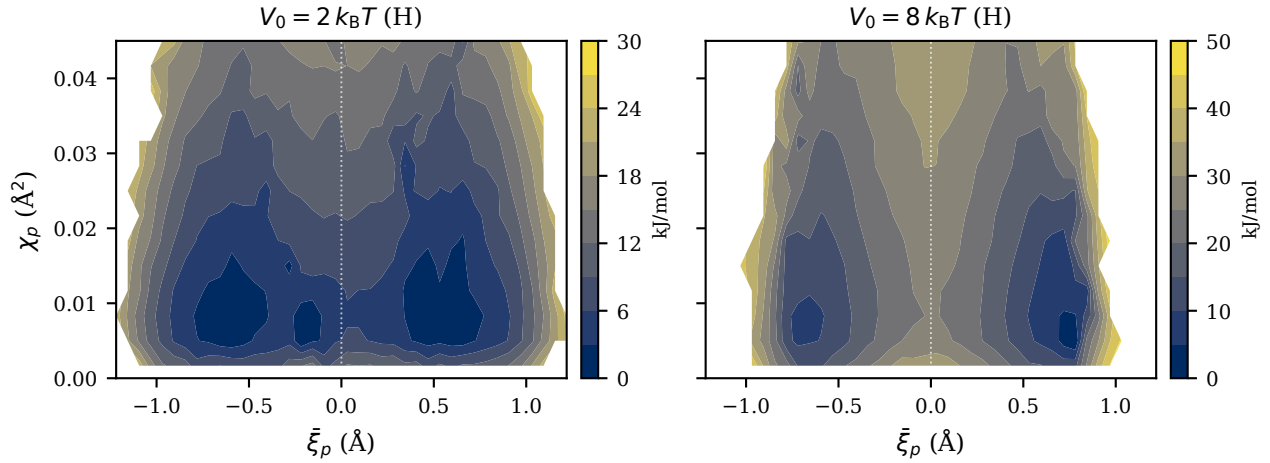


FIG. 5. WHAM free energy $F(\bar{\xi}_p, \chi_p)$ for the asymmetric double well (H, $T = 300 \text{ K}$, $P = 32$). The vertical line is the centroid dividing surface.

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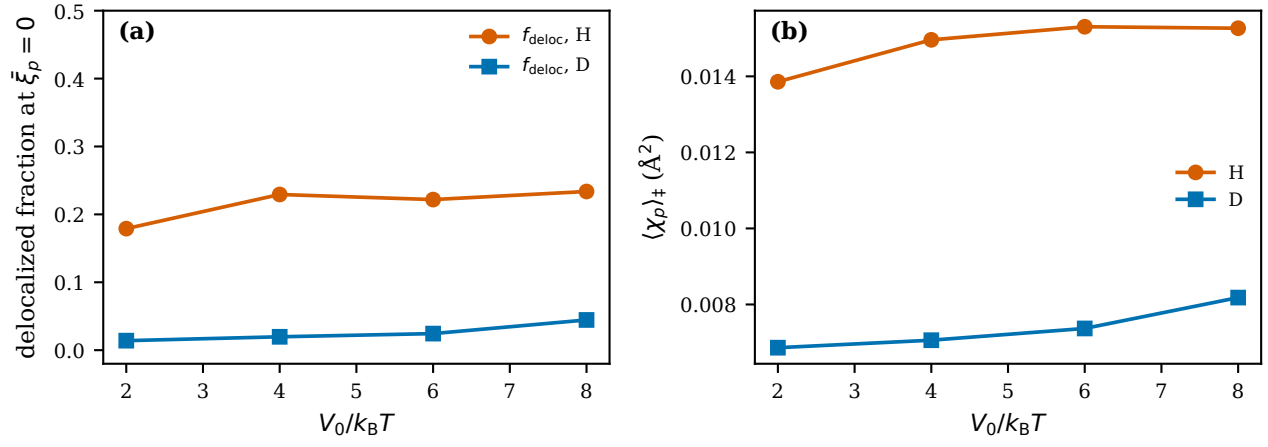


FIG. 6. ADW dividing-surface statistics versus barrier height. (a) Delocalized-path fraction f_{deloc} for H and D. (b) Mean path spread $\langle \chi_p \rangle_{\ddagger}$.

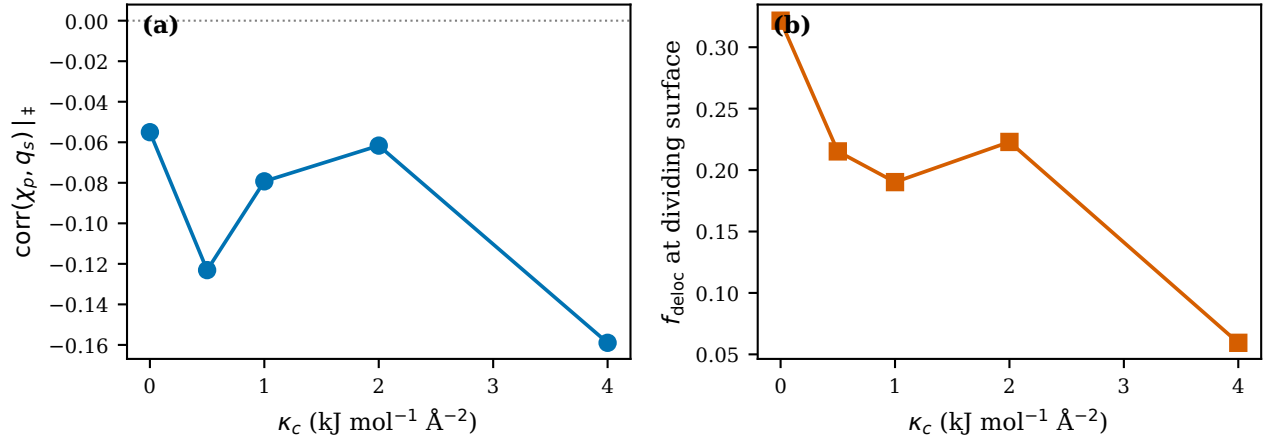


FIG. 7. Two-dimensional proton-solvent model. (a) Dividing-surface correlation of path spread with the classical solvent coordinate. (b) Delocalized fraction versus coupling.